



# **LECTURE 1**

## **INTRODUCTION TO MANUFACTURING**

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1. Objectives
2. Manufacturing
3. Competitiveness in manufacturing
4. Computer Integrated Manufacturing(CIM)
5. CIM components
6. Conclusion



# Objectives

1. Formulate CIM-specific concepts
2. Understanding the competitiveness
3. Describe the CIM characteristics
4. Characterize CIM components





# Manufacturing



*Manufacturing is a collection of interrelated activities that includes product design and documentation, material selection, planning, production, quality assurance, management, and marketing of goods.*

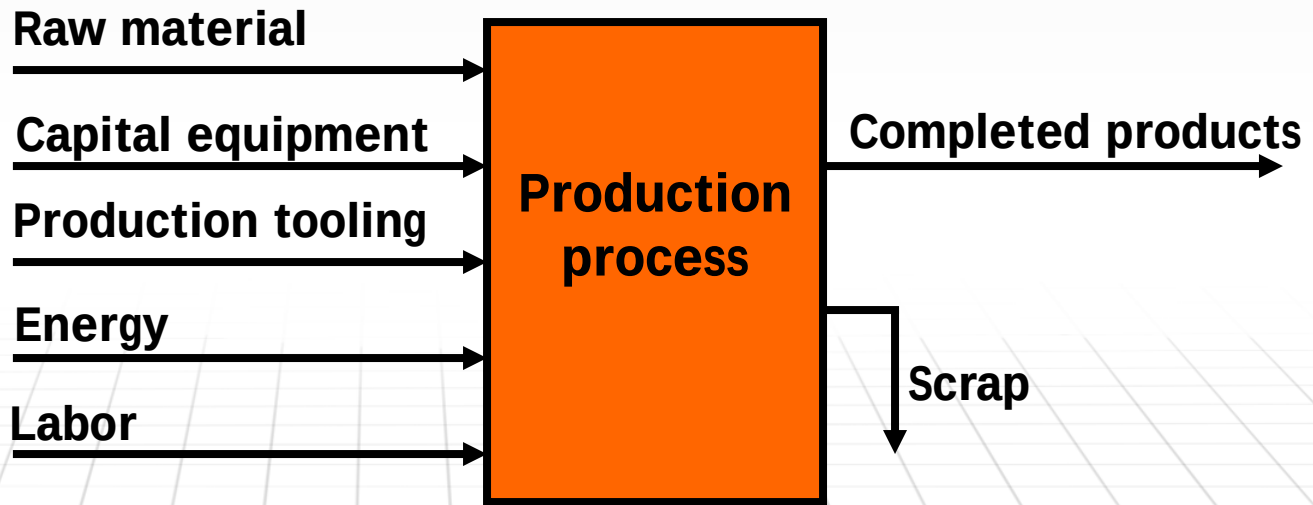
*Computer Integrated Manufacturing (CIM) involves the integration of production systems using integrated systems and data communication correlated to management strategies that improve the organization and efficiency of the human resources.*



# The goal



The fundamental goal of manufacturing is to use the activities (design and documentation, material selection, planning, production, quality assurance, management, and marketing of goods) to convert raw materials into finished goods on a profitable basis.



*Manufacturing model*



**Competitiveness  
(1)**



**CHALLENGES**



**EXTERNAL**

**INTERNAL**

## Competitiveness (2)

Internet



Costs



Global economy



New/niche  
entrants



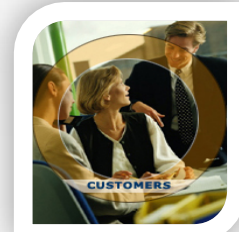
Traditional  
competition



Suppliers



Partnerships and  
alliances



Customers

External challenges



## Competitiveness (3)

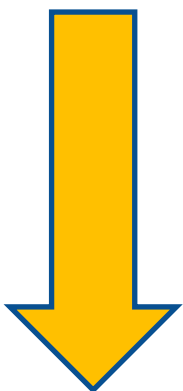
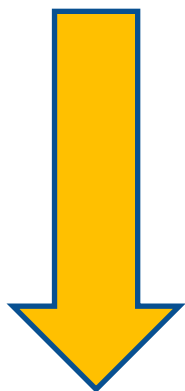
The **internal challenge** is to develop a manufacturing strategy to minimize the negative impact of the external challenges on the success of the business.

*A manufacturing strategy is a plan or process that forces congruence between the corporate objectives and the marketing goals, and the production capability of a company.*

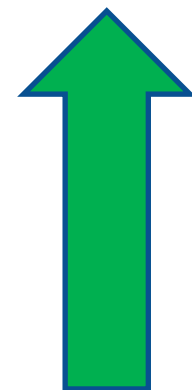
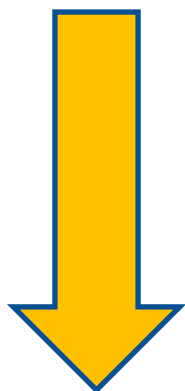


Computer integrated manufacturing must ensure:

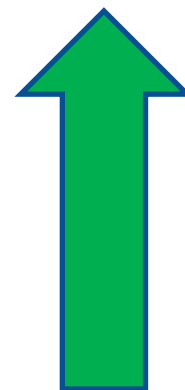
Lower production costs  
Reducing the size of production lots



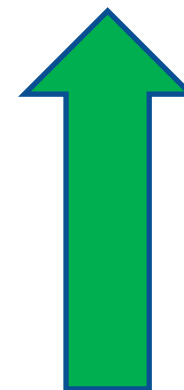
Inventory reduction



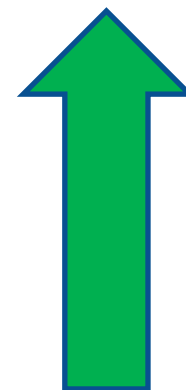
Better control of the production process



Increasing flexibility



Better relationship with customers



Improved quality





**Factors that  
characterize a  
CIM system**

- Setup time;
- Quality;
- Manufacturing-space ratio;
- Inventory;
- Flexibility;
- Distance;
- Uptime.



# Setup time

*Setup time is the time required to get a machine ready for production.*

The burden rate is a cost roll-up or total that includes:

- Recovery of the original cost of the process machine and associated support hardware;
- Maintenance costs (fault and prevention);
- Labor cost (direct and indirect);
- Programming cost for CNC machines and robots.



# Self-assessement test



1. Partnerships and alliances are external challenges for a company.

**T**

**F**

2. The results of a production process are not only finite products.

**T**

**F**

3. In CIM, manufacturing batch size reduction leads to increased production costs.

**T**

**F**

4. The increase of the setup time increases the productivity of a manufacturing system.

**T**

**F**





# Quality

*Quality is expressed as a percentage of defective parts produced or as a percentage of total sales.*

*Warranty quality-defective parts discovered after the product is shipped to the customer.*

*Total quality represents the total cost in percentage of sales that quality costs the enterprise*

Captured quality-defects found before the product is shipped to the customer, has units of parts per million (ppm).



# Manufacturing-space ratio

*The manufacturing-space ratio standard is a measure of how efficiently manufacturing is utilized.*

*The total footprint of the machines, plus the area of workstations where value is added to the product, is divided by the total manufacturing space.*



Value is added at any workstation that processes the part or assembles the part with another part.

The larger this ratio, the more efficient the production operation.

The manufacturing-space ratio must be greater than 50% according to the world-class standard.



# Inventory

*Inventory is represented by raw materials, partially completed parts, subassemblies, and finished goods that are waiting to be shipped to the customer.*

## Inventory

**movement**

work-in-process inventories

**organization**

**cycle stock**

**safety stock**

**anticipation  
stock**



# Inventory

**Inventory levels** are measured in terms of **inventory turns**, or velocity, and in **residence time**.

The number of **inventory turns** for a product is equal to the annual COSt of Goods Sold (COGS) divided by the average inventory value.

The **residence time** is the average number of days that a part spends in production.



The goal is to reduce the **inventory levels** => number of inventory turns increases => residence time decreases.



## Flexibility

*Flexibility is measured as the number of different parts that can be produced on the same machine. Flexibility also measures the ability to produce new product designs in a short time.*



## Distance

*The standard distance measures the total linear meters of a part's travel through the plant from receiving the raw material to finished products in shipping.*



## Uptime

*Uptime is the percentage of time a machine is producing to specifications compared to the total time that production can be scheduled.*



# Self-assessment test

1. The increase of flexibility in a manufacturing system, involves, usually, additional cost.

**T**

**F**

2. The anticipation stock is not a safety stock.

**T**

**F**

3. The smaller the space-manufacturing ratio, the more efficient the manufacturing system is.

**T**

**F**

4. The cycle stock exists when the sales are larger than the production.

**T**

**F**

5. The production is smaller if a machine availability is larger.

**T**

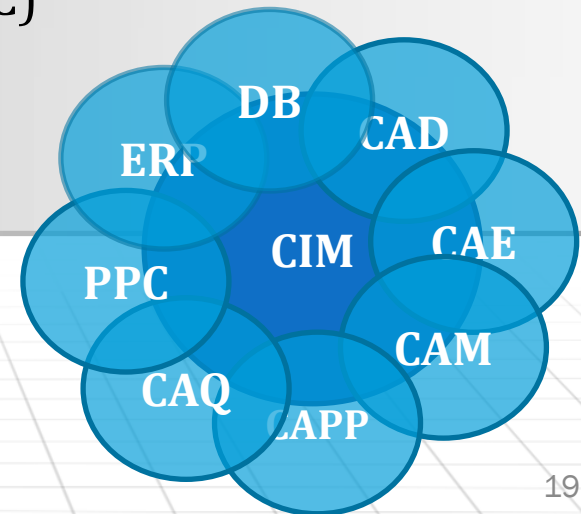
**F**



# CIM components

CIM systems include several subsystems whose efficiency determine the manufacturing system performance:

- Computer-Aided Design (CAD)
- Computer-Aided Engineering (CAE)
- Computer-Aided Manufacturing (CAM)
- Computer-Aided Process Planning (CAPP)
- Computer-Aided Quality assurance (CAQ)
- Production Planning and Control (PPC)
- Enterprise Resource Planning (ERP)
- A business system integrated by a common database (DB)

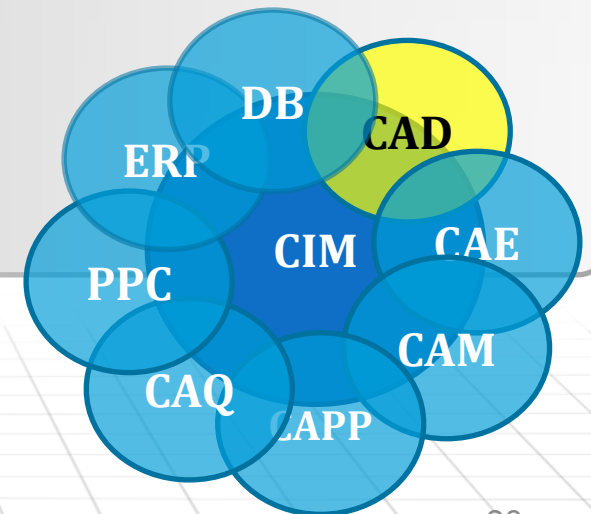


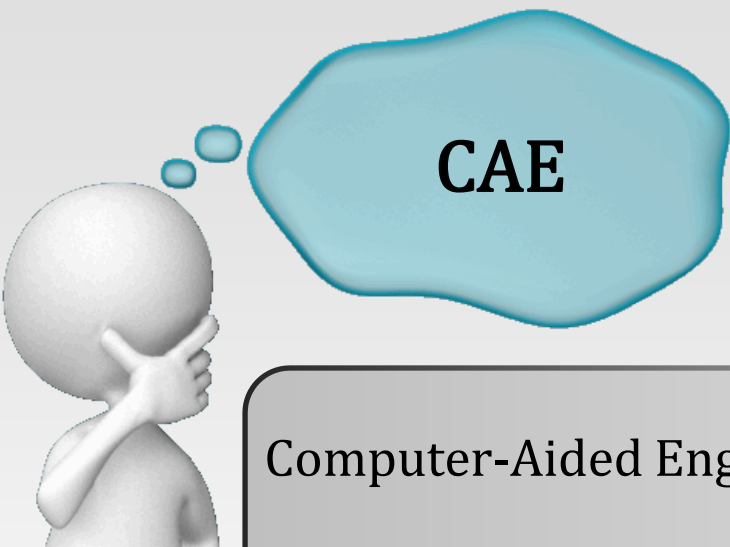


*Computer-Aided Design (CAD) is defined as the use of computer systems to assist in the creation, modification, analysis and documentation of a part design.*

The use of CAD assures:

- ✓ Reduced design time
- ✓ Improved design quality
- ✓ Improved documentation
- ✓ Creation of a database

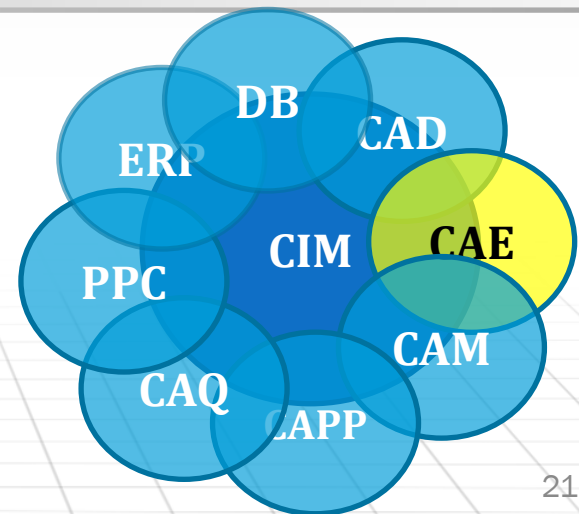


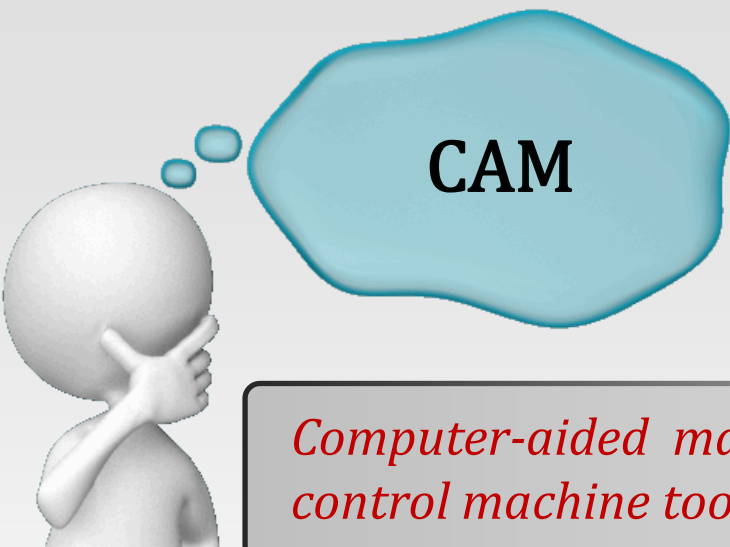


# CAE

Computer-Aided Engineering is used for:

- ✓ Stress analysis in components and assemblies,
- ✓ Kinematics,
- ✓ Simulation and analysis of processes as casting, molding, and die press forming,
- ✓ Optimization of the product or process.



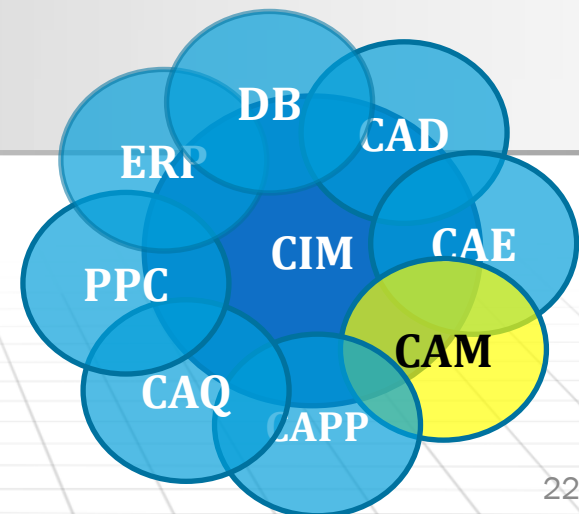


# CAM

*Computer-aided manufacturing is the use of computer software to control machine tools and related machinery in manufacturing.*

Equipment in a CIM:

- Industrial robots,
- Computer numerical controlled machine tools,
- Transport systems
- Automated warehouses, etc.



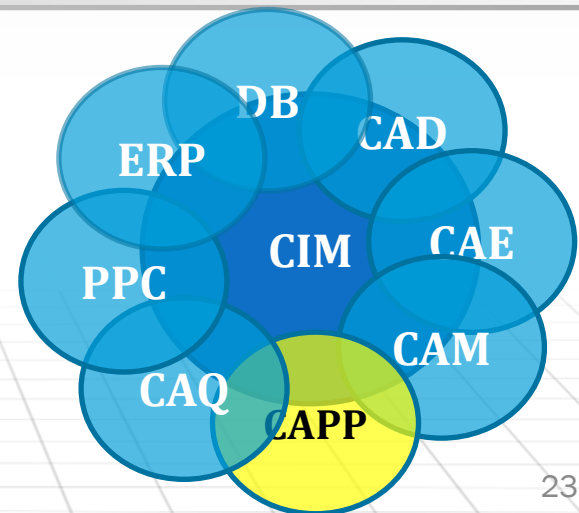


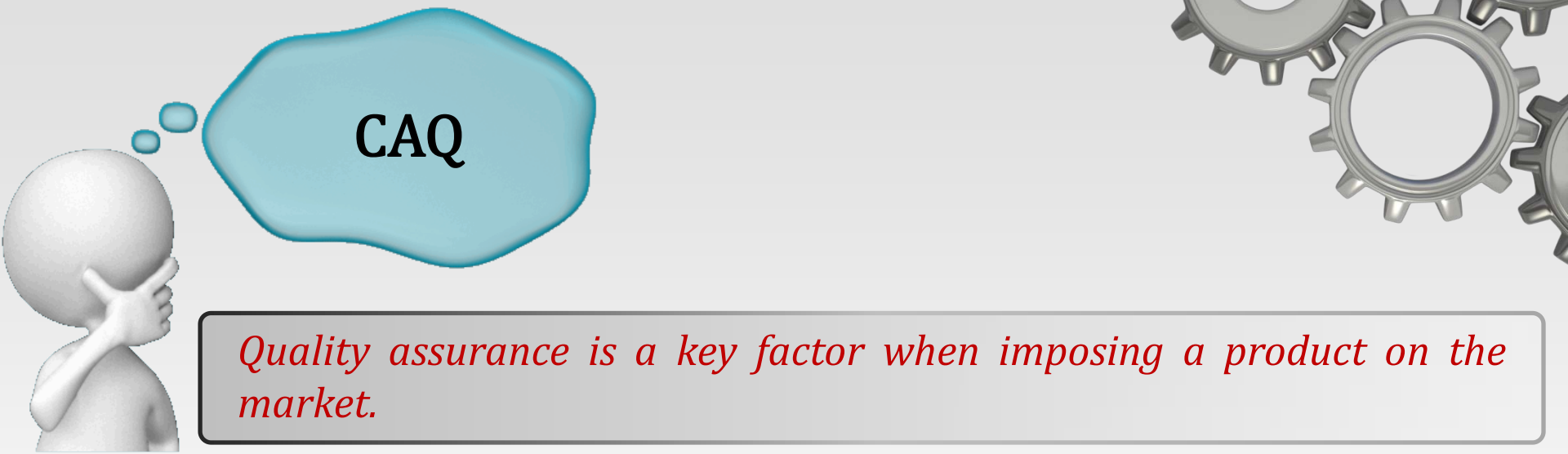
# CAPP

*Computer-Aided Process Planning, ensures a direct connection between design and production.*

CAPP main effects:

- Increase flexibility,
- Decrease planning time
- Decrease human intervention.

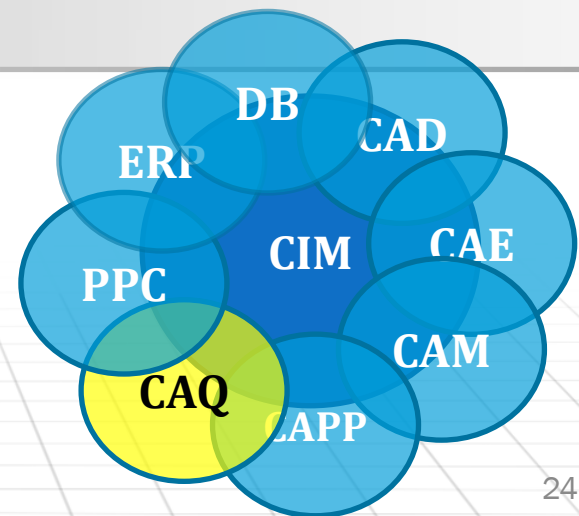




*Quality assurance is a key factor when imposing a product on the market.*

In general, implementation of quality control techniques leads to additional costs.

Early detection of faults may reduce significantly the production costs.





# Self-assessment test

1. CAD increases design time.

**T**

**F**

2. CAE is used in various systems simulation.

**T**

**F**

3. CAM uses the computer for several processes.

**T**

**F**

4. CAPP decreases the process planning time.

**T**

**F**



# PPC

*Production Planning and Control (PPC) concerns the aspects of manufacturing logistics.*

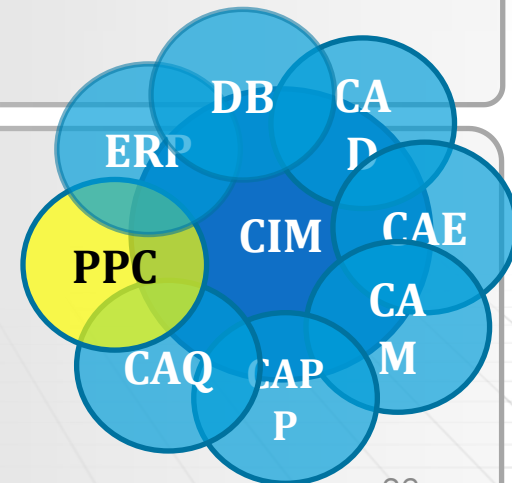
*Production control follows the way production resources are allocated and, if necessary, takes appropriate decisions to correct any deviations from the required production plan.*

Production control involves:

- Control of production lines,
- Inventory control, etc.

Production planning:

- Determine the products to be manufactured,
- How much of which will be produced,
- Production deadlines,
- Planning of labor and necessary equipment

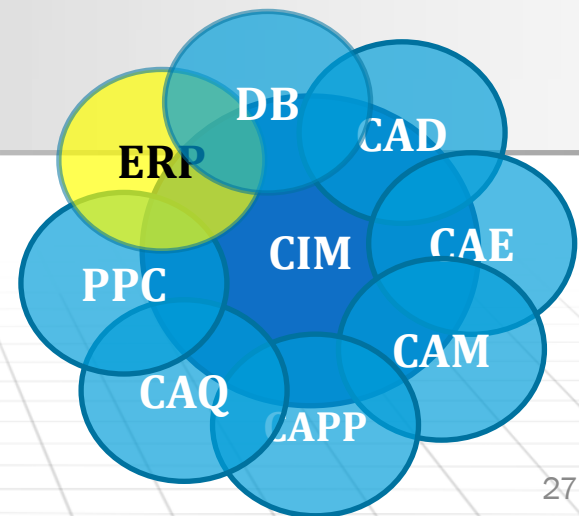


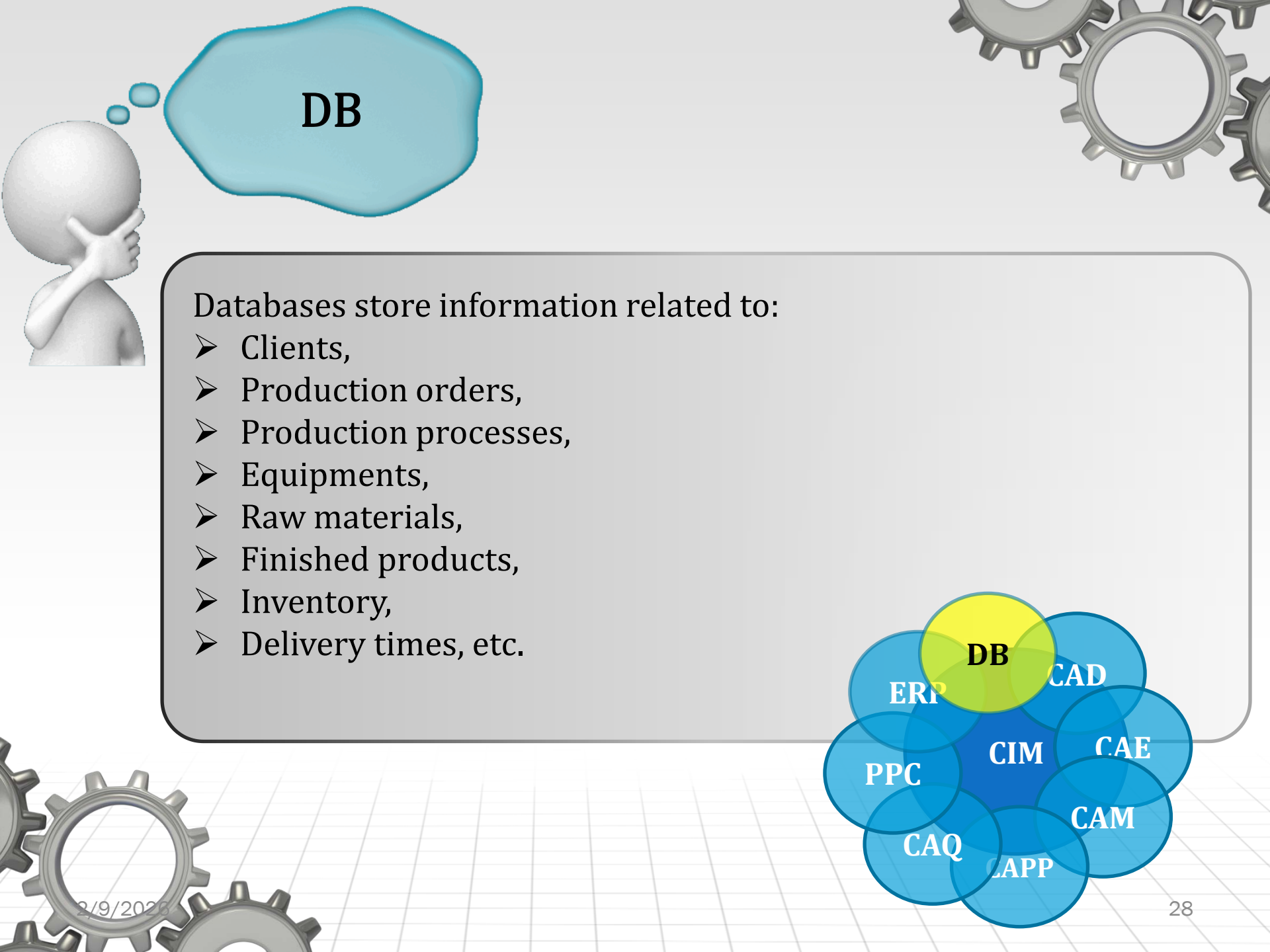


# ERP

*Enterprise Resource Planning integrates information from various departments:*

- *Financial accounting,*
- *Production,*
- *Sales,*
- *Service,*
- *Customer relations, etc.*

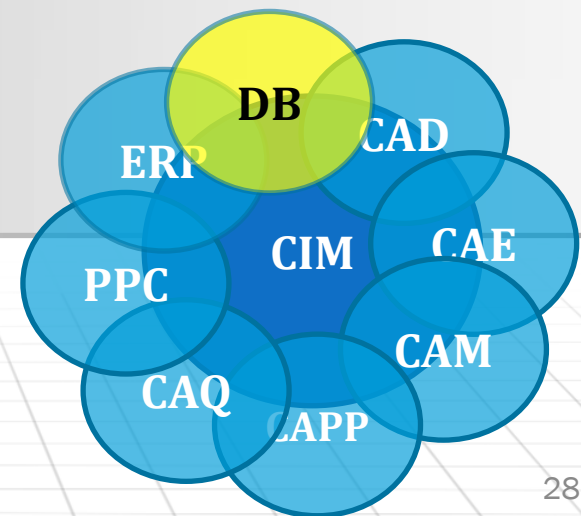




**DB**

Databases store information related to:

- Clients,
- Production orders,
- Production processes,
- Equipments,
- Raw materials,
- Finished products,
- Inventory,
- Delivery times, etc.





# Self-assessment test

1. Quality assessment is more effective if done during the production cycle.

**T**

**F**

2. Production control involves also the inventory control.

**T**

**F**

3. Production planning does not involve the labor planning.

**T**

**F**

4. Enterprise resource planning (ERP) is efficient in CIM.

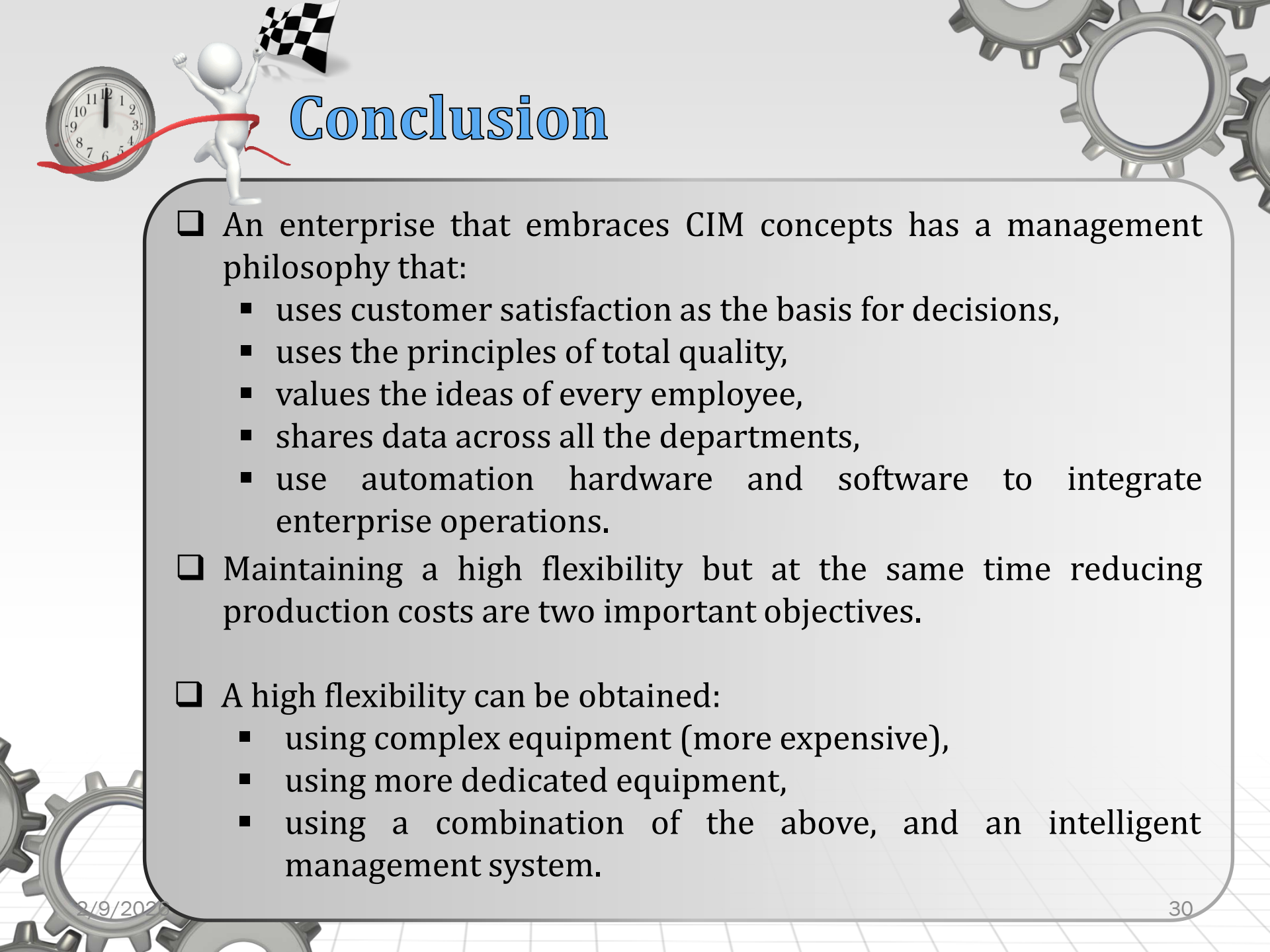
**T**

**F**

5. Databases are the binding agents between different departments of a company.

**T**

**F**



# Conclusion

- ❑ An enterprise that embraces CIM concepts has a management philosophy that:
  - uses customer satisfaction as the basis for decisions,
  - uses the principles of total quality,
  - values the ideas of every employee,
  - shares data across all the departments,
  - use automation hardware and software to integrate enterprise operations.
- ❑ Maintaining a high flexibility but at the same time reducing production costs are two important objectives.
- ❑ A high flexibility can be obtained:
  - using complex equipment (more expensive),
  - using more dedicated equipment,
  - using a combination of the above, and an intelligent management system.



- ☐ The components of a manufacturing system.
- ☐ Types of manufacturing systems.
- ☐ CIM classification factors.
- ☐ CIM Progress functions.